

TCS NQT — Top 50 Coding Questions

C++ Solutions | Based on the uploaded TCS NQT coding-question PDF

All 50 questions converted to C++ in the same order as the source document.

1. Check Even or Odd

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;
    cout << (n % 2 == 0 ? "Even" : "Odd");
    return 0;
}
```

2. Check Prime Number

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    bool prime = true;
    if (n <= 1) prime = false;

    for (int i = 2; i <= sqrt(n); i++) {
        if (n % i == 0) {
            prime = false;
            break;
        }
    }

    cout << (prime ? "Prime" : "Not Prime");
    return 0;
}
```

3. Factorial of a Number

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    long long fact = 1;
    for (int i = 1; i <= n; i++)
        fact *= i;

    cout << fact;
    return 0;
}
```

4. Fibonacci Series — First N Terms

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int a = 0, b = 1;
    for (int i = 1; i <= n; i++) {
        cout << a << " ";
        int c = a + b;
        a = b;
        b = c;
    }
}
```

```

    return 0;
}

```

5. Reverse a Number

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int rev = 0;
    while (n != 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }

    cout << rev;
    return 0;
}

```

6. Check Palindrome Number

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int temp = n, rev = 0;
    while (n != 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }

    cout << (temp == rev ? "Palindrome" : "Not Palindrome");
    return 0;
}

```

7. Armstrong Number

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int temp = n, sum = 0;
    while (n != 0) {
        int digit = n % 10;
        sum += digit * digit * digit;
        n /= 10;
    }

    cout << (temp == sum ? "Armstrong" : "Not Armstrong");
    return 0;
}

```

8. Sum of Digits

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int sum = 0;
    while (n != 0) {
        sum += n % 10;
        n /= 10;
    }

    cout << sum;
    return 0;
}

```

9. Largest of Three Numbers

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int a, b, c;
    cin >> a >> b >> c;

    cout << max(a, max(b, c));
    return 0;
}
```

10. GCD of Two Numbers

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int a, b;
    cin >> a >> b;

    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }

    cout << a;
    return 0;
}
```

11. LCM of Two Numbers

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int a, b;
    cin >> a >> b;

    int x = a, y = b;
    while (y != 0) {
        int temp = y;
        y = x % y;
        x = temp;
    }

    int gcd = x;
    int lcm = (a * b) / gcd;
    cout << lcm;

    return 0;
}
```

12. Check Leap Year

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int year;
    cin >> year;

    if ((year % 4 == 0 && year % 100 != 0) ||
        (year % 400 == 0))
        cout << "Leap Year";
    else
        cout << "Not Leap Year";

    return 0;
}
```

13. Count Vowels and Consonants

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);

    int vowels = 0, consonants = 0;

    for (char ch : str) {
        ch = tolower(ch);
        if (isalpha(ch)) {
            if (string("aeiou").find(ch) != string::npos)
                vowels++;
            else
                consonants++;
        }
    }

    cout << "Vowels: " << vowels << endl;
    cout << "Consonants: " << consonants;
    return 0;
}
```

14. Reverse a String

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);

    string rev = "";
    for (int i = str.length() - 1; i >= 0; i--)
        rev += str[i];
}
```

```

    cout << rev;
    return 0;
}

```

15. Check Anagram

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string s1, s2;
    getline(cin, s1);
    getline(cin, s2);

    sort(s1.begin(), s1.end());
    sort(s2.begin(), s2.end());

    cout << (s1 == s2 ? "Anagram" : "Not Anagram");
    return 0;
}

```

16. Remove Duplicates from String

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);

    string result = "";
    for (char ch : str) {
        if (result.find(ch) == string::npos)
            result += ch;
    }

    cout << result;
    return 0;
}

```

17. Find Second Largest in Array

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int first = INT_MIN, second = INT_MIN;

    for (int num : arr) {
        if (num > first) {
            second = first;
            first = num;
        } else if (num > second && num != first) {
            second = num;
        }
    }

    cout << second;
    return 0;
}

```

18. Linear Search

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int key;

```

```

    cin >> key;

    bool found = false;
    for (int num : arr) {
        if (num == key) {
            found = true;
            break;
        }
    }

    cout << (found ? "Found" : "Not Found");
    return 0;
}

```

19. Binary Search — Sorted Array

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int key;
    cin >> key;

    int low = 0, high = n - 1;
    bool found = false;

    while (low <= high) {
        int mid = (low + high) / 2;

        if (arr[mid] == key) {
            found = true;
            break;
        } else if (arr[mid] < key) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }

    cout << (found ? "Found" : "Not Found");
    return 0;
}

```

20. Bubble Sort

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    for (int i = 0; i < n - 1; i++)
        for (int j = 0; j < n - i - 1; j++)
            if (arr[j] > arr[j + 1])
                swap(arr[j], arr[j + 1]);

    for (int num : arr) cout << num << " ";
    return 0;
}

```

21. Selection Sort

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    for (int i = 0; i < n - 1; i++) {
        int minIndex = i;

        for (int j = i + 1; j < n; j++)
            if (arr[j] < arr[minIndex])
                minIndex = j;

        swap(arr[i], arr[minIndex]);
    }

    for (int num : arr) cout << num << " ";
    return 0;
}
```

22. Insertion Sort

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    for (int i = 1; i < n; i++) {
        int key = arr[i], j = i - 1;

        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = key;
    }

    for (int num : arr) cout << num << " ";
    return 0;
}
```

23. Matrix Addition

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int r, c;
    cin >> r >> c;

    vector<vector<int>> a(r, vector<int>(c));
    vector<vector<int>> b(r, vector<int>(c));

    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            cin >> a[i][j];

    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            cin >> b[i][j];

    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++)
            cout << a[i][j] + b[i][j] << " ";
        cout << endl;
    }

    return 0;
}
```

```
}
```

24. Transpose of Matrix

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int r, c;
    cin >> r >> c;

    vector<vector<int>> a(r, vector<int>(c));

    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            cin >> a[i][j];

    for (int j = 0; j < c; j++) {
        for (int i = 0; i < r; i++)
            cout << a[i][j] << " ";
        cout << endl;
    }

    return 0;
}
```

25. Count Frequency of Element in Array

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int key;
    cin >> key;

    int count = 0;
    for (int num : arr)
        if (num == key) count++;

    cout << count;
    return 0;
}
```

26. Check if Array is Sorted

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    bool sorted = true;

    for (int i = 0; i < n - 1; i++) {
        if (arr[i] > arr[i + 1]) {
            sorted = false;
            break;
        }
    }

    cout << (sorted ? "Sorted" : "Not Sorted");
    return 0;
}
```

27. Merge Two Arrays

```
#include <bits/stdc++.h>
using namespace std;

int main() {
```

```

int n1;
cin >> n1;

vector<int> a(n1);
for (int i = 0; i < n1; i++) cin >> a[i];

int n2;
cin >> n2;

vector<int> b(n2);
for (int i = 0; i < n2; i++) cin >> b[i];

for (int num : a) cout << num << " ";
for (int num : b) cout << num << " ";

return 0;
}

```

28. Find Missing Number — 1 to N

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n - 1);
    for (int i = 0; i < n - 1; i++) cin >> arr[i];

    int total = n * (n + 1) / 2;
    int sum = 0;

    for (int num : arr) sum += num;

    cout << total - sum;
    return 0;
}

```

29. Count Words in a String

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);

    stringstream ss(str);
    string word;
    int count = 0;

    while (ss >> word) count++;

    cout << count;
    return 0;
}

```

30. Remove All Spaces from String

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);

    string result = "";

    for (char ch : str)
        if (ch != ' ') result += ch;

    cout << result;
    return 0;
}

```

31. Find Duplicate Elements in Array

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                cout << arr[i] << " ";
                break;
            }
        }
    }

    return 0;
}
```

32. Move All Zeros to End

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int index = 0;

    for (int i = 0; i < n; i++)
        if (arr[i] != 0)
            arr[index++] = arr[i];

    while (index < n)
        arr[index++] = 0;

    for (int num : arr) cout << num << " ";
    return 0;
}
```

33. Rotate Array Right by 1 Position

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int last = arr[n - 1];

    for (int i = n - 1; i > 0; i--)
        arr[i] = arr[i - 1];

    arr[0] = last;

    for (int num : arr) cout << num << " ";
    return 0;
}
```

34. Check Palindrome String

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);
```

```

    string rev = "";

    for (int i = str.length() - 1; i >= 0; i--)
        rev += str[i];

    cout << (str == rev ? "Palindrome" : "Not Palindrome");
    return 0;
}

```

35. Count Number of Digits

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int count = 0;

    while (n != 0) {
        n /= 10;
        count++;
    }

    cout << count;
    return 0;
}

```

36. Sum of Elements in Array

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    int sum = 0;

    for (int i = 0; i < n; i++) {
        cin >> arr[i];
        sum += arr[i];
    }

    cout << sum;
    return 0;
}

```

37. Find Minimum Element in Array

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int minimum = arr[0];

    for (int i = 1; i < n; i++)
        if (arr[i] < minimum)
            minimum = arr[i];

    cout << minimum;
    return 0;
}

```

38. Pattern Printing — Right Triangle

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;
}

```

```

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++)
            cout << "**";
        cout << endl;
    }

    return 0;
}

```

39. Power of a Number

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int base, exp;
    cin >> base >> exp;

    int result = 1;

    for (int i = 1; i <= exp; i++)
        result *= base;

    cout << result;
    return 0;
}

```

40. Decimal to Binary

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    string binary = "";

    while (n > 0) {
        binary = char('0' + n % 2) + binary;
        n /= 2;
    }

    cout << binary;
    return 0;
}

```

41. Binary to Decimal

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    string binary;
    cin >> binary;

    int decimal = 0;
    int power = 0;

    for (int i = binary.length() - 1; i >= 0; i--) {
        if (binary[i] == '1')
            decimal += pow(2, power);
        power++;
    }

    cout << decimal;
    return 0;
}
```

42. Check Perfect Number

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int sum = 0;

    for (int i = 1; i <= n / 2; i++)
        if (n % i == 0)
            sum += i;

    cout << (sum == n ? "Perfect" : "Not Perfect");
    return 0;
}
```

43. Strong Number

```
#include <bits/stdc++.h>
using namespace std;

int factorial(int n) {
    int fact = 1;
    for (int i = 1; i <= n; i++)
        fact *= i;
    return fact;
}

int main() {
    int n;
    cin >> n;

    int temp = n, sum = 0;

    while (n != 0) {
        int digit = n % 10;
        sum += factorial(digit);
        n /= 10;
    }

    cout << (temp == sum ? "Strong" : "Not Strong");
    return 0;
}
```

44. Count Even and Odd Numbers in Array

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    int even = 0, odd = 0;
```

```

    for (int i = 0; i < n; i++) {
        int x;
        cin >> x;

        if (x % 2 == 0)
            even++;
        else
            odd++;
    }

    cout << "Even: " << even << endl;
    cout << "Odd: " << odd;

    return 0;
}

```

45. Find Intersection of Two Arrays

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n1;
    cin >> n1;

    vector<int> a(n1);
    for (int i = 0; i < n1; i++) cin >> a[i];

    int n2;
    cin >> n2;

    vector<int> b(n2);
    for (int i = 0; i < n2; i++) cin >> b[i];

    for (int i = 0; i < n1; i++) {
        for (int j = 0; j < n2; j++) {
            if (a[i] == b[j]) {
                cout << a[i] << " ";
                break;
            }
        }
    }

    return 0;
}

```

46. Check Substring

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string str, sub;

    getline(cin, str);
    getline(cin, sub);

    if (str.find(sub) != string::npos)
        cout << "Substring Present";
    else
        cout << "Substring Not Present";

    return 0;
}

```

47. Remove Specific Character from String

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);

    char ch;
    cin >> ch;

    string result = "";

    for (char c : str)

```

```

        if (c != ch)
            result += c;

    cout << result;
    return 0;
}

```

48. Sum of Prime Numbers up to N

```

#include <bits/stdc++.h>
using namespace std;

bool isPrime(int n) {
    if (n <= 1) return false;

    for (int i = 2; i <= sqrt(n); i++)
        if (n % i == 0)
            return false;

    return true;
}

int main() {
    int n;
    cin >> n;

    int sum = 0;

    for (int i = 2; i <= n; i++)
        if (isPrime(i))
            sum += i;

    cout << sum;
    return 0;
}

```

49. Reverse Words in a Sentence

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string str;
    getline(cin, str);

    stringstream ss(str);
    vector<string> words;
    string word;

    while (ss >> word)
        words.push_back(word);

    for (int i = words.size() - 1; i >= 0; i--)
        cout << words[i] << " ";

    return 0;
}

```

50. Two Sum Problem

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) cin >> arr[i];

    int target;
    cin >> target;

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] + arr[j] == target) {
                cout << i << " " << j;
                return 0;
            }
        }
    }
}

```

```
}  
  
cout << "No Pair Found";  
return 0;  
}
```